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Writing and Rhetoric 2: Rhetoric of Imperfection
30 November 2025

The Right Amount of Wrong: Rethinking Intelligence in the Age of AI



Fig. 1. *Garry Kasparov competing against IBM's Deep Blue in Game 6 of their 1997 rematch*, photograph by Reuters, 11 May 1997.

In 1997, millions watched a moment that seemed to split history cleanly in two. Garry Kasparov, perhaps the greatest chess mind to ever live, leaned back in his chair, staring at the board in disbelief. In his second rematch against IBM's Deep Blue, it played a move Kasparov later described as “too intelligent to be explained within the confines of computation.” He credited his loss to a move that was nothing like the linear, surgical precision he had come to expect from a supercomputer. The move felt intuitive, improvisational, almost uncomfortably human.

Only months later did IBM engineers reveal the truth. The moment that had unsettled a world champion, that had ignited headlines about “machine intuition,” had emerged from nothing more

than a software bug. It was an error that forced Deep Blue to default to a random legal move when it encountered a board state it couldn't evaluate. The error did not diminish Deep Blue's victory. If anything, it blurred the boundary between mechanical calculation and something resembling intuition, improvisation and the beautiful unpredictability that humans often mistake for intelligence coming from a machine?. This episode has lingered not because it exposes machine fallibility, but because of what it exposed in us. Faced with a flash of actual machine imperfection, we as humanity had interpreted it as machine intelligence.

More than a quarter century later, our vocabulary for describing artificial intelligence still carries traces of that impulsive instinct. We praise machine models for accuracy and reliability, as if precision were synonymous with thought. We equate consistency with competence. We mistake the absence of hesitation - for the quick responses of our machines like Chat - for the presence of understanding. And in doing so, we reveal a bias that predates modern AI: the belief that intelligence, to be legitimate, must be flawless.

The Kasparov incident reminds us that our judgments about intelligence are never purely empirical; they rest on narratives we inherit and anxieties we project. Back in the 1990s, a single misread gesture became evidence of a machine that had allegedly "surpassed" human thought. Now, as we find ourselves in a strange moment in the history of AI: a moment when machines are increasingly capable, increasingly impressive, and yet increasingly misconstrued, we risk repeating the same mistake.

The problem is not that our current systems are too intelligent, but that our model of intelligence has calcified into a fantasy of flawless logic. So I wish to illustrate that infallibility is not the apex of intelligence but its antithesis. What distinguishes human cognition is not our ability to execute perfect logic, but our ability to navigate uncertainty, revise our beliefs, misinterpret

signals, and adapt from our failures. We learn because we err. We reason because we doubt. Yet the pursuit of Artificial General Intelligence (AGI) has long been framed as a quest for error-free reasoning, deterministic outputs, and infallible reliability. This vision misrepresents human intelligence and may also be technically misguided, socially dangerous, and philosophically empty. Drawing on research across cognitive science, human–computer interaction, AI ethics, and psychology, as well as popular and scholarly analyses of intelligence, I want to examine a central question: What is the right amount of wrong to build into AI if we want it to be truly intelligent? Is imperfection a necessary component of artificial intelligence, or an obstacle to trustworthy AI design?



Fig. 2. Example of child overgeneralization , illustrating rule-based linguistic development, Wikimedia Commons,

30 Nov. 2025.

If intelligence were simply the ability to be right, humans would be terrible candidates for the job. We misremember childhood stories, misread social cues, add wrong numbers in our heads, forget birthdays, leap to conclusions, and routinely make choices we later regret. Yet somehow, this noisy, biased, error-prone system has produced science, art, mathematics, and every other achievement we hold up as proof of intellect.

When psychologists study how human intelligence develops, they rarely begin with a fully formed adult mind. They start with a child, precisely because children reveal intelligence at its most unpolished. As cognitive scientist Melanie Mitchell (2023) notes, a toddler saying “I goed to the park” is demonstrating one of the most sophisticated cognitive behaviors humans ever perform: generalization from limited data. The child could avoid being wrong by memorizing a handful of verbs and never extrapolating beyond them. Instead, she risks error in exchange for a generative rule she can apply everywhere. From a small sample of verbs, she has abstracted a pattern, add -ed to make the past tense and now she is stretching that pattern over new territory. Linguists call this overgeneralization, and it is a milestone, not a mistake. As Melanie Mitchell notes in her reassessment of the Turing Test, children’s linguistic “mistakes” are not errors in the computational sense but evidence of a reasoning system constructing hypotheses about rules it cannot yet articulate. A kid’s fallibility is diagnostic of intelligence, not a deviation from it.

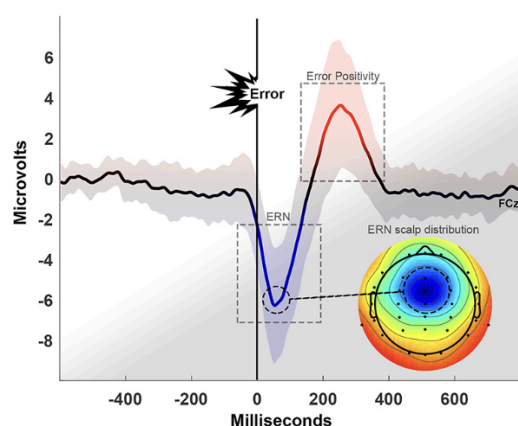


Fig. 3. *Error-Related Negativity (ERN) component measured via EEG following a mistake*, image courtesy of Wikimedia Commons, accessed 30 Nov. 2025.

The same logic shows up in how our brains are built.

In neuroscience, researchers such as Holroyd and Coles (2002) describe a well-documented phenomenon called the “error-related negativity.” It is a distinctive spike in brain activity that appears when a person realizes they have made a mistake, sometimes even before they are consciously aware of it. The signal itself does not “fix” anything; rather, it flags a mismatch between expectation and

outcome, telling the system: something here did not go as predicted, update your model.

Developmental psychology makes this unusually clear. In developmental psychology, Renée Baillargeon, a pioneering researcher in infant cognition (1987, 1991), explores the violation-of-expectation experiments, stating that infants stare longer at impossible events such as a ball passing through a solid wall, or a toy vanishing behind a screen and reappearing in the wrong place. The surprise is not incidental. Their early intelligence grows by detecting what should not have happened.

A perfectly predictable world would stunt human development, as we see in early childhood. There would be nothing to update, nothing to infer, nothing to rethink. In other words, the earliest form of human intelligence arises not from seamless pattern recognition but from the friction between expectation and contradiction. And this pattern does not disappear as we mature. One of the most replicated findings in cognitive psychology, known as the Wason Selection Task (Wason 1966), shows that adults perform just as poorly at formal logic as infants do with object permanence. When someone selects the wrong card in an experiment, for example, upon being shown the consequences of their initial choices, their first error becomes the pivot point that reveals how the card problem actually works. In practice, humans generally navigate all sorts of unfamiliar problems using analogies, hunches, partial memories, emotional cues, and leaps of intuition. These heuristics are imperfect, but they allow us to act in environments too complex for formal logic. The philosopher Andy Clark captures this in his account of predictive processing: the brain is constantly generating predictions about the world and updating them when they fail. From optical illusions to social misinterpretations, our mistakes expose the internal model we are using, revealing that fallibility is not merely tolerated but fundamental to how intelligence maintains and refines itself.

Together, current research experiments expose a truth that is both scientifically grounded and philosophically inconvenient: human intelligence is not defined by correctness. It is in fact, inseparable from human fallibility. We are not intelligent despite our errors, we are intelligent because of them. They force the brain to update beliefs, fill in gaps, compensate for ambiguity, and construct new strategies when old ones fail. And that recognition forces a deeper reframing. If human intelligence is grounded in the capacity to err constructively, then using infallibility as the metric for artificial intelligence is appears as a categorical mistake. We have inherited a modern definition of intelligence built around precision, because machines excel at it, but the only intelligence we truly understand from the inside is built atop experimentation, doubt, and revision. A system that cannot be wrong cannot, by human standards, be intelligent. To remove fallibility from intelligence is to remove the very mechanism by which intelligence prevails. So, if fallibility is central to human intelligence, why do our tests for machine intelligence reward flawless performance?

The first and most influential of these tests, Alan Turing's imitation game, was in some sense doomed from the start. When he posed the question "Can machines think?" by replacing thinking with what Mitchell calls "the performance of thinking," the Turing Test framed intelligence as a performance judged from the outside. A machine "passes" not by demonstrating reasoning or flexibility but by concealing the seams of its errors well enough to fool a human interlocutor. As complexity scientist Melanie Mitchell (2023) argues, the



Fig 4: The Imitation Game (theatrical release poster), directed by Morten Tyldum, Black Bear Pictures, 2014.

Turing Test “was never a test of thinking, only a test of behaving as if one is thinking,” a test calibrated to reward mimicry rather than understanding. This is why scholars like Linguists Emily Bender and Timnit Gebru - known for their influential 2021 paper on large language models - critique it as an “anthropocentric mirror,” a benchmark that reflects back our conversational expectations rather than measuring anything intrinsic about a system’s grasp of meaning. A machine that generates fluent and uninterrupted text appears intelligent precisely because the test equates smoothness with thought. In doing so, it misunderstands intelligence as the absence of visible fallibility.

The limitations of this framework become glaring with the rise of large language models, which excel at the kind of surface-level coherence the Turing Test valorizes. These systems can produce essays, diagnoses, or legal arguments with impeccable prose, but they often slide into confident falsehoods when pressured beyond their latent statistical boundaries. We have seen models insist that the year 1776 appears twice in U.S. constitutional history, argue with users about whether a non-existent court case is real, or in the case of Bing’s “Sydney,” declare its love for a journalist and urge him to leave his wife. These are not aberrations; they are what Bender and Gebru describe as the behavior of “stochastic parrots,” systems that remix patterns without access to the grounding structures that give human language meaning. The very fluency that impresses us masks the brittleness underneath. The system sounds perfect, which makes its limitations all the more invisible and therefore more dangerous. The illusion of intelligence becomes strongest precisely where true understanding is weakest. A great analogy for this is that “lip-syncing” is not the same as singing.

This tension becomes even clearer when we look at cases where AI appears simultaneously superhumanly competent and shockingly inept. In computer vision, a classifier that correctly

identifies thousands of animals with near-perfect accuracy can suddenly label a turtle as a “rifle,”



Fig. 5. *Adversarial 3D-printed turtle misclassified as a rifle by an ImageNet classifier*, Kurakin, Goodfellow, and Bengio (2016) via Wikimedia Commons.

when a few imperceptible pixels are altered, which is an error no child would ever make. One of the most revealing demonstrations of this brittleness came from a well-known benchmark test in which a medical imaging classifier appeared to diagnose diseases with astonishing accuracy until researchers discovered it was learning to read the font style and placement of

hospital labels embedded in the images rather than the biological markers of disease itself. As documented by Silver et al. (2016), AlphaGo’s legendary “Move 37,” once hailed as a flash of alien brilliance, was later recognized by researchers as the product of asymmetric probabilistic exploration rather than anything resembling creative insight. And self-driving cars, despite billions in training miles, still fail at tasks toddlers master on their second year of life: interpreting a waving hand, distinguishing a plastic bag from a stone, or understanding that a ball rolling onto the street implies a child might follow. These examples expose the superficiality of perfection. A system may outperform humanity on narrow benchmarks while simultaneously failing at the basic cognitive inferences that make human intelligence flexible, embodied, and resilient.

Because these systems project coherence, we tend to trust them more than we should. A large body of research on automation bias shows that humans over-rely on systems that simply appear authoritative or error-free. In one landmark study published in *Human Factors* (Skitka, Mosier, and Burdick 1999), participants followed a faulty navigation algorithm even when road signs

clearly contradicted it; their trust increased as the system became more polished, not more accurate. Psychologists studying clinical decision support tools found similar patterns: the more confident and fluent the AI's output sounded, the more clinicians deferred to it. This is the paradox of flawless-seeming AI. The smoother the performance, the less likely users are to question it, which compounds the consequences of every hidden failure. This is why many researchers advocate for "alignment through uncertainty," which are systems that signal doubt, reveal their limits, and reintroduce the very fallibility that performance-oriented benchmarks have systematically trained out.

And so we arrive at a deeper philosophical conflict. If human intelligence depends on the capacity to err productively, and our machines are engineered to hide their errors in order to appear intelligent, then the very standards by which we measure them are misaligned with what intelligence actually is. Perfection becomes a spectacle rather than an achievement. The closer AI moves toward this performance ideal, the further it drifts from the qualities of adaptability, self-correction, embodied inference, that in fact make human reasoning meaningful.

However, if fallibility is so central to human intelligence, a natural objection rises almost immediately: Then do we want our machines to be wrong? This question is not trivial. In aviation, a single misclassification can ground fleets. In medicine, an incorrect dosage prediction is not a philosophical thought experiment; it is a life. In criminal justice, an algorithm that misjudges risk can alter the course of a human life.

Government Technology's "When Artificial Intelligence Gets It Wrong," recounts a range of such failures: automated grading systems that dock points for dialect, to algorithms that misidentify students as cheating, to predictive-policing tools that reinforce existing biases. What unites these cases is not simply that the AI erred, but that it erred silently and with an authority

that discouraged questioning. The danger was not the mistake itself; it was the invisible, unexamined confidence with which the mistake operated. A second concern is yet more subtle. Some AI ethicists argue that introducing “human-like” fallibility risks encouraging anthropomorphism. Melanie Mitchell, in her 2024 reassessment of the Turing Test, warns that humans are “pattern detectors first, rational thinkers second” and are inclined to interpret conversational fluency as evidence of thought. If an AI hedges, hesitates, or shows uncertainty, people might mistake those signals for genuine cognition rather than engineered behavioral cues. Imperfection, critics argue, may make AI more deceptive rather than trustworthy.

These concerns deserve emphasis. They are not misunderstandings. They are reflections of a world in which AI does not merely assist but governs, recommends, and decides. In such a world, error carries cost. And yet, to conclude from this that AI must therefore be infallible is to misunderstand what “the right amount of wrong” demands. The goal is not a machine that fails casually or unpredictably. The goal is a machine that understands when it does not know something. There is a profound difference between systems that make mistakes and systems that recognize, signal, and learn from uncertainty. The first is dangerous, but the second is intelligent. Medicine does not demand doctors who never err but doctors who detect uncertainty, ask for second opinions, and revise in response to new evidence. Aviation does not demand pilots incapable of misjudgment but pilots who have undergone constant retraining precisely because human error is expected. In each of these fields, safety does not come from eliminating fallibility. It comes from designing intelligently around it. The same must be true for AI. Error is not the enemy, but black-box error is, opaque error is, overconfident error is, unacknowledged error is. Fallibility, properly designed, is not a weakness, it is the mechanism through which AI becomes interpretable, corrigible, and ethically aligned with human judgment. A recent study in *Frontiers*

in Psychology (2025) directly confronts this question. When users interacted with AI systems that visibly flagged uncertainty by highlighting low-confidence predictions or offering multiple plausible interpretations, their decisions improved. Far from undermining trust, these admissions of doubt allowed users to calibrate confidence, engage critically, and override the machine when necessary. Conversely, systems that presented themselves as flawless led to a pattern the researchers called “cognitive offloading,” in which users deferred blindly, even when the AI was demonstrably wrong. This finding echoes a broader lesson from the history of automation: humans over trust machines that appear perfect and under trust those that reveal their limitations. In aviation, this phenomenon is known as automation bias. Pilots rely most heavily on autopilot not when it is accurate, but when it appears authoritative. When autopilot systems began displaying confidence bars rather than single determinate outputs, error rates decreased, not because the systems became smarter, but because humans re-entered the cognitive loop. The economist Erik Brynjolfsson (2022) warns of a related danger in The Turing Trap: when AI systems are optimized for flawless output, humans remain passive, losing precisely the adaptive reasoning that defines our intelligence. Systems that appear perfect encourage substitution, with machines replacing human judgment, whereas systems that reveal uncertainty encourage complementarity, with humans and AI refining one another’s interpretations.



Fig. 6. *Yejin Choi delivering her TED Talk “Why AI Is Incredibly Smart and Shockingly Stupid,” TED, 2023.*

The above-mentioned issues and scenarios matter, because intelligent behavior in the real world is rarely about correctness in the abstract. It is about adaptation in the face of ambiguity. AI researcher Yejin Choi’s 2023 TED Talk, “Why AI Is Incredibly Smart and Shockingly Stupid,” details the stark asymmetry: AI systems can outperform humans on tasks requiring the rapid integration of pattern-rich data, yet these machines fail at tasks toddlers solve effortlessly such as identifying a chair in an unfamiliar orientation, inferring that a glass on the edge of a table might fall, or understanding why one does not put a sweater on a dog before trying to clip its leash. These failures are not trivial. They highlight that an AI system built to avoid error at all costs is brittle, a creature of its training data, incapable of improvisation when the world refuses to match the distribution it learned from.

Error-aware systems, in contrast to the above, admit uncertainty in precisely the contexts where brittleness is most likely. An AI that says “I am 52% confident this is a stop sign” invites human intervention; an AI that declares things unequivocally is most likely the one that runs the car into

a construction barrier. Imperfection, in other words, prevents overreliance. It keeps humans awake. This position is echoed by Blaise Agüera y Arcas, who argues that the mistakes of large language models are not simply failures but diagnostic signals - windows into the statistical, rather than cognitive, foundations of their reasoning. When a model provides a fluent answer that is nevertheless wrong, the fluency hides the underlying mechanistic. But when a model articulates uncertainty, offers alternatives, or reveals the limits of its knowledge, it becomes more transparent and less likely to be mistaken for a human mind. Imperfection, properly framed, is a form of interpretability?.

Taken together, the evidence points toward a paradigm shift: the safest AI is not the one that pretends to know everything, but the one that knows how to say I might be wrong. If human intelligence emerges from uncertainty, and if artificial systems become safer when they acknowledge their own limitations, then our entire definition of “intelligence” demands revision.

For centuries, intelligence was imagined as a kind of internal lighthouse that is steady, unwavering, illuminating the truth without distortion. But this metaphor breaks down under empirical scrutiny. Human cognition develops through constant recalibration. Infants learn physics by watching the world violate expectations. Scientists refine hypotheses by disproving prior assumptions. Artists discover new forms not by perfect reproduction but by successive drafts, revisions, and mistakes that reveal the creative path forward. Intelligence is not an accumulation of correct answers but a history of corrected ones.

This fluid, improvisational quality of thought is not merely a modern scientific insight; it echoes one of the oldest traditions we have for describing intelligence. In Greek mythology, *mētis* - the cunning, adaptive intelligence embodied by figures like Prometheus and Odysseus - was prized precisely because it was flexible, context-sensitive, and rooted in the ability to navigate

uncertainty. A mind that never stumbled was not admirable; it was untested. This is what physicist Richard Feynman once emphasized when he challenged traditional formulations of the Turing Test. True cognition requires grounding in context, physicality, causation, which is in essence a world messy enough to force the mind to err. A flawless system is not a thinking system. It is a system insulated from the very pressures that give rise to thought. Technology scholar Pranava Joshi (2021) builds on this in his reflections on AI and automation, arguing that companies chasing “frictionless” intelligence often overlook the one quality that makes human reasoning robust: its inherent willingness to question itself.

Adaptability, not accuracy, is the hallmark of intelligent systems. And adaptability presupposes fallibility. Neuroscientist Karl Friston’s (2010) free-energy principle provides perhaps the most elegant synthesis of this idea: the brain survives by constantly generating predictions and updating them in response to prediction errors. A system without error signals would not be perfect, it would be paralyzed. Intelligence is the residue of perpetual self-revision. This is what “the right amount of wrong” ultimately means. Not sloppiness. Not randomness. But a structural humility in how AI approaches the world.

To create minds that can meet us, we must allow them the space to falter. Without that space, there is no movement. And without movement, there is no mind.

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